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Characterization of ethylene oxide–propylene oxide block copolymers by combination of different chromatographic techniques and matrix-assisted laser desorption ionization time-of-flight mass spectroscopy

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ABSTRACT

Block copolymers of ethylene oxide (EO) and propylene oxide (PO) are characterized by combination of two-dimensional chromatography and MALDI-TOF-MS. Liquid chromatography under critical conditions (LCCC) is used as first dimension and fractions are collected, mobile phase evaporated and diluted in the mobile phase used in second dimension (SEC or LAC). This two-dimensional chromatography in combination of MALDI-TOF-MS gives information about purity of reaction products, presence of the byproducts, chemical composition and molar mass distribution of all the products.

1. Introduction

Block copolymers of ethylene oxide (EO) and propylene oxide (PO) are in widespread use due to their amphiphilic nature [1–4]. Triblocks of ethylene oxide and propylene oxide with PO as internal block (EO–PO–EO) are called poloxamers. They are commercially available under different trade names, such as Pluronic and Synperonic [5,6]. Triblocks with the EO block in the center (PO–EO–PO) are also commercially available. Some studies on EO–PO diblocks have also been reported [7,8].

In anionic polymerization of propylene oxide, there are significant side reactions [9]: chain transfer to the monomer (Fig. 1) leads to poly(oxypropylene) chains with an allyl end group [10–13]. Allylic end groups may also be formed by a reaction with the penultimate unit. This reaction produces 1,2-propandiol, which might act as initiator resulting in the formation of PPGs (Fig. 2).

In the presence of water, polypropylene glycols (PPGs) may also be formed. These side reactions are not easy to avoid completely, but can be minimized by carefully selecting reaction conditions [7].

The length of these homopolymers may be very different from that of the PPO block in the copolymer, which becomes clear from the following considerations. The PPG chains originate from traces of water present in the initial reaction mixture, hence they start growing at the very beginning of the polymerization. As they have two hydroxy end groups, they may grow in both directions. Consequently, they should be approximately twice as large as the PPO block in the diblock copolymer. The PPO chains with allylic end group, however, result from a chain transfer reaction, which happens all the time, which means, that some chains will be started at rather high conversion. As they have much less time to grow, they will be shorter than the PPO block in the copolymer. The reaction with the penultimate unit will also happen all the time, hence there may be PO-diols formed at rather high monomer conversion. Consequently, they should have a much lower molecular weight (as these chains have less time to grow) than those resulting from initial water content.

Of course the amount of the homopolymers formed by these reactions, their functionality and molar mass distribution (MMD) can enormously affect the properties of the product. So there is a strong need for reliable analytical methods, which can yield the required information, namely the molar mass distribution, the amount and molar mass of the homopolymers, the length of the individual blocks in the copolymer and their arrangement (diblocks or triblocks) as well as the sequence (EO–PO–EO or PO–EO–PO). Of

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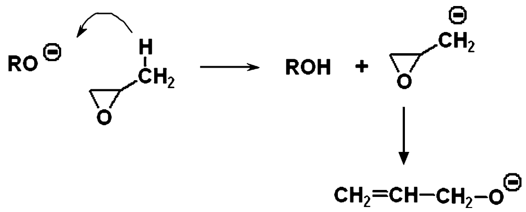

Fig. 1. Formation of monoallyl-PPOs by chain transfer to the monomer.

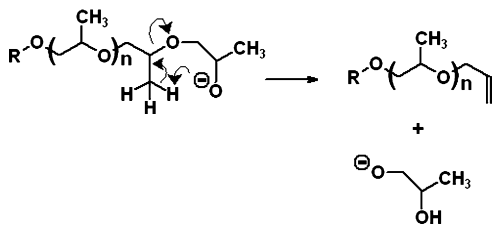

Fig. 2. Formation of monoallyl-PPOs and PPGs by reaction with the penultimate unit.

course this can only be achieved by combination of different separation methods, typically by two-dimensional liquid chromatography (2D-LC) [14–16].

Pasch and co-workers have shown that two-dimensional chromatography (2D-LC) can be used for many different types of amphiphilic systems. They used 2D-LC as well as combinations of HPLC with MALDI-TOF-MS, FTIR-spectroscopy and NMR [17–21].

In 2D-LC, most studies used liquid chromatography under critical conditions (LCCC) in the first and SEC in the second dimension, but other combinations may also be useful.

In previous papers we have described the independent determination of the length of the EO and PO blocks [22,23]. In another communication we have reported critical conditions for poly(alkylene oxides) on various columns in different mobile phases [24] and studied the elution behaviour of other structural units under these conditions.

The aim of the present study was the independent determination of the functionality and molar mass distribution of the homopolymers and the PO block in the copolymers. The MMD of the EO block can be determined by analysis of the PEG or PEG-MME used as starting material.

For this purpose, LCCC of PPO should be used in the first dimension, which allows a separation according to functionality. This should also be performed on a semipreparative scale, which allows an analysis of each fraction in the second dimension by different chromatographic techniques as well as by MALDI-TOF-MS.

2. HPLC of polymers

The separation mechanism in liquid chromatography of polymers depends on the size/pore size-ratio and on the so-called interaction parameter c, which describes the interaction of the structural unit with stationary phase [25]. This parameter is negative in size exclusion chromatography (SEC) in which retention decreases with increase in molar mass and is positive in liquid adsorption chromatography (LAC) where retention increases exponentially with the number of repeat units.

In both cases, the elution volume V_e depends upon the distribution coefficient K of the polymer between the interstitial volume V_i and the pore volume V_p:

$$V_e = V_i + K \cdot V_p \qquad (1)$$

This equation is well known in SEC (where K can assume values between 0 and 1): a polymer of very high molar mass (above the exclusion limit) elutes at V_i (as $K = 0$), while small molecules (which have access to the entire pore volume) elute at the void volume $V_0 = V_i + V_p$ (as $K = 1$).

In LAC, the distribution coefficient can assume very high values (much larger than 1). The distribution coefficient is related to the free energy change related to the transition of the polymer from the interstitial volume into the pores of the stationary phase:

$$\ln K = \Delta G = \Delta H - T \cdot \Delta S \qquad (2)$$

In general, the driving force in SEC is the entropy change, while in LAC retention is governed by the enthalpic interaction of the repeat unit with the stationary phase.

For a given system (polymer–stationary phase–mobile phase) there may be a mobile phase composition (and temperature), where entropic and enthalpic contributions compensate each other ($\Delta H = T\Delta S$), hence $\Delta G = 0$ and $K = 1$.

The transition point between SEC and LAC is often termed as "critical conditions" or critical adsorption point (CAP). At the CAP, the interaction parameter has value of zero: under such conditions, a non-functional homopolymer elutes – independent of its molar mass – at the void volume of the column. In other words, the polymer chain becomes chromatographically invisible [26–29], hence it is possible to separate a complex polymer according to other structural units (different from the repeat unit), such as end groups or other blocks.

These three modes of liquid chromatography can be combined in the analysis of functional polymers and copolymers by two-dimensional chromatography.

In the case of block copolymers the interaction parameters c_A and c_B of the individual structural units (A and B) may have different values. In a mobile phase, where both c_A and c_B are strongly negative, the molar mass distribution (MMD) can be determined by SEC, with retention decreasing with increasing molar mass. In SEC, the influence of chemical composition is often rather small. SEC yields the MMD of the entire molecule (provided that the calibration functions for A and B are comparable). With dual detection [30–32] one may obtain information on the chemical composition across the MMD.

SEC is, however, not capable of discriminating mixtures of homopolymers with different repeat unit or polymers with the same repeat unit, but different functionality. This can be achieved by LCCC.

At the CAP of the repeat unit A, the elution volume is independent of the chain length of the block A. In this case, AB copolymers can be separated according to the block B.

Basically, there are two possible situations, which may be utilized:

1. If the interaction parameter c_B of the block B is negative, the block copolymer will elute earlier than the homopolymer A (in SEC order). Obviously, situation 1 will only be useful, if the block B is sufficiently large: it can be used for separation of homopolymers from block copolymers [7,27,33–37].
2. If c_B is positive, the block copolymer elutes later than the homopolymer A (in LAC order) [22,23]. Situation 2 is limited to short blocks in the case of strong interaction (i.e. large c_B). It can be very useful, if the non-critical block is monodisperse [36,38].

3. Experimental

EO–PO diblock copolymers were synthesized by microwave assisited anionic ring opening polymerization of PO by using PEG-MME as initiator as has been reported elsewhere [7]. It has been shown, that there are side reactions in anionic ring opening polymerization of propylene oxide leading to PPO homopolymers with allylic end group and PPGs . In this study we used 2D-LC by using

LCCC in first dimension and SEC, LAC and MALDI-TOF-MS in second dimension for complete analysis of all fractions.

In SEC, we used a modular system consisting of a Gynkotek 300C pump, a VICI injector equipped with a 100 μl sample loop, two column selection valves Rheodyne 7060 (Rheodyne, Cotati, CA, USA), a density detection system DDS 70 (Chromtech, Graz, Austria) coupled with a Bischoff 8110 refractive index detector (Bischoff, Leonberg, Germany). SEC measurements were performed on a 600 mm × 7.8 mm Phenogel 100 Å column (Phenomenex, Torrance, CA, USA) or a 300 mm × 7.8 mm PLgel MIXED E column (Polymer Laboratories, Church Stretton, Shropshire, UK) at a flow rate of 1.00 ml/min. Sample concentrations in case of solid products were 3.0–10.0 g/l while in case of fractions appropriate volumes were injected to get reasonable signal. The columns were calibrated with PPGs from Sigma–Aldrich and FLUKA (Buchs, Switzerland). Molar mass distributions were calculated using CHROMA.

For analytical scale LCCC, the mobile phase was delivered by an ISCO 2350 HPLC pump (ISCO, Lincoln, NE, USA) at a flow rate of 0.5 ml/min. Samples were injected using an autosampler Spark SPH 125 Fix (Spark Holland, Emmen, The Netherlands) equipped with a 20 μl loop. A refractive index detector RI 2414 (Waters) or a SEDEX 45 ELSD (Sedere, France) was connected to the DDS 70. Nitrogen was used as carrier gas, and the pressure at the nebulizer was set to 1.0 bar. Evaporator temperature: 30 °C.

Reversed phase column used for LCCC on analytical scale was Prodigy ODS3 (Phenomenex, Torrance, CA, USA): silica-based octadecyl phase; 250 mm × 4.6 mm; particle diameter = 5 μm; nominal pore size = 100 Å.

For semipreparative scale chromatography, the mobile phase was delivered by JASCO 880 PU pump (Japan Spectrosopic Company, Tokyo, Japan) at a flow rate of 2.0 ml/min. Samples were injected manually using Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a 500 μl loop. A semipreparative column was used for all measurements: Spherisorb ODS2 (300 mm × 25 mm, 5 μm, 80 Å) from PhaseSep (Deeside, UK). The separations by LCCC in the first dimension were performed using the density detection system DDS 70 (CHROMTECH, Graz, Austria) which has been developed by our group. Data acquisition and processing was carried out by using software package CHROMA. A SICON LCD 201 RI detector was coupled to the DDS70. Fractions were collected using an Advantec SF 2120 fraction collector (Advantec, Dublin, CA, USA).

The solvent was removed from the fractions by freeze drying using a Benchtop 2K ES (Virits USA). The limits of the instrument are −60 °C and 2.6 Pa. Samples were frozen to about −40 °C and were subjected for drying to benchtop 2K ES which was premaintained at temperature below −50 °C and pressure below 3 Pa. Typically it took 2–3 h for complete drying of samples.

In LAC, the separations in the second dimension were performed on a gradient system Ultimate 3000, consisting of a pump DGP-3600A, solvent degasser SRD-3600, column thermostat TCC-3000, autosampler WPS-3000SL, all from Dionex (Germering, Germany), an evaporative light scattering detector PL-ELS 2100 (Polymer Laboratories, Church Stretton, Shropshire, UK). Data acquisition and processing was performed using the software Chromeleon (Dionex, Germering, Germany). Gradient separations were performed on a Symmetry C18 5 μm column (from Waters, Milford, MA, USA: silica-based octadecyl phase; 150 mm × 3.0 mm; particle diameter: 5 μm, pore size = 100 Å) at a flow rate of 0.5 ml/min. The injected sample volumes depended on the total volume of the fraction from the first dimension in order to inject comparable sample sizes.

The following gradient was used: start 70 wt% methanol–water, in 35 min to 90 wt% methanol, 9 min constant 90%, then back to 70% methanol in 1 min.

PPGs were purchased from Sigma–Aldrich. The solvents for HPLC (THF, chloroform, methanol and water, all HPLC grade) were purchased from Roth (Karlsruhe, Germany). Mobile phases were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a measuring cell DMA 602 M (A. Paar, Graz, Austria).

Sample concentrations in SEC, LAC and analytical LCCC were 3–10 g/l, in the fractionation 10–15 g/l.

The columns and density cells were placed in a thermostatted box, in which a constant temperature (25.0 °C for LCCC and at 30.0 °C for SEC) was maintained for all measurements using a thermostat Lauda RM6, Lauda-Königshofen, Germany).

MALDI-TOF mass spectrometry was performed on a Micromass TofSpec 2E Time-of-Flight Mass Spectrometer. The instrument is equipped with a nitrogen laser (337 nm wavelength, operated at a frequency of 5 Hz), and a time lag focusing unit. Ions were generated by irradiation slightly above the threshold laser power. Positive ion spectra were recorded in reflectron mode applying an accelerating voltage of 20 kV and externally calibrated with a suitable mixture of poly(ethyleneglycol)s (PEGs). The spectra of 100–150 shots were averaged to improve the signal-to-noise ratio. Analysis of data was done with MassLynx-Software V3.5 (Micromass/Waters, Manchester, UK).

Sample was prepared by mixing a solution of dithranol (c = 10 mg/mL in THF), a solution of the analyte (c = 2 mg/mL in THF), and a solution of CF_3COONa (NaTFA; c = 0.1 mg/mL in THF), in a ratio of 7:1:1 (v/v/v). 0.5 μl of the resulting mixture were deposited on the sample plate (stainless steel) and allowed to dry under air.

4. Results and discussion

The first question concerns the presence of homopolymers in the copolymer. As mentioned above, the PPOs with allylic end group should have a much lower molar mass than the block copolymer, hence it should appear as a second peak or at least as a shoulder in SEC. Using coupled density and RI detection, the composition across the MMD can be determined [30,39,40]: when a mass m_i of a copolymer containing the weight fractions w_A and w_B $(= 1 - w_A)$ of the structural units A and B elutes in the interval i of the peak, its composition (w_A) is obtained from the areas $x_{i,j}$ of the slice in both detectors (denoted by j), and the corresponding response factors $f_{j,A}$ and $f_{j,B}$:

$$\frac{1}{w_A} = 1 - \frac{((x_{i,1}/x_{i,2}) \times f_{2,A} - f_{1,A})}{((x_{i,1}/x_{i,2}) \times f_{2,B} - f_{1,B})} \tag{3}$$

Once the composition of a fraction is known, its mass can be calculated:

$$m_i = \frac{x_{i,1}}{w_A \times (f_{1,A} - f_{1,B}) + f_{1,B}} \tag{4}$$

A prerequisite of this technique is a linear dependence of the response factors on chemical composition, which has been verified in previous papers [30–32]. The molar mass dependence of the response factors can be accounted for by the software.

As can be seen in Fig. 3, there is indeed a low molecular fraction, which obviously consists of homopolymer of PO. The main peak contains 40% PO, which is reasonable, as the sample should contain on average 50% PO, and a considerable amount of the PO has gone to the homopolymer fraction. It is, however, not clear, whether the main peak contains just the block copolymer or also some amount of PPGs, which may have a similar molar mass.

This can only be shown by LCCC at the CAP for PPO. As we have shown in a previous paper [7], a CAP for the hydrophobic PPO block can be found on reversed phase columns in THF–water mobile phases containing about 80% THF. Under such conditions, the hydrophilic EO block elutes in the exclusion regime, i.e. earlier than the solvent peak, and so does the block copolymer. Moreover,

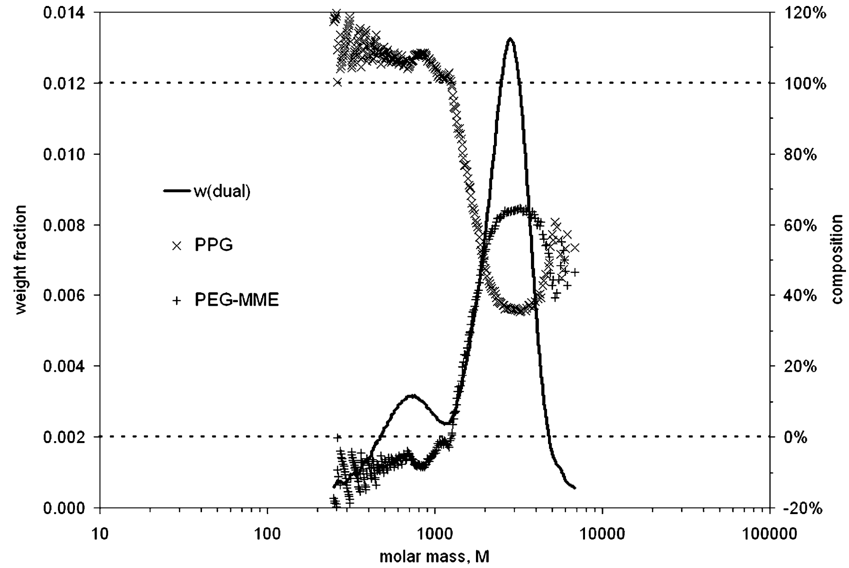

Fig. 3. MMD and chemical composition of MeO-PEG 2000-b-PPO (raw sample) as obtained by SEC with dual detection.

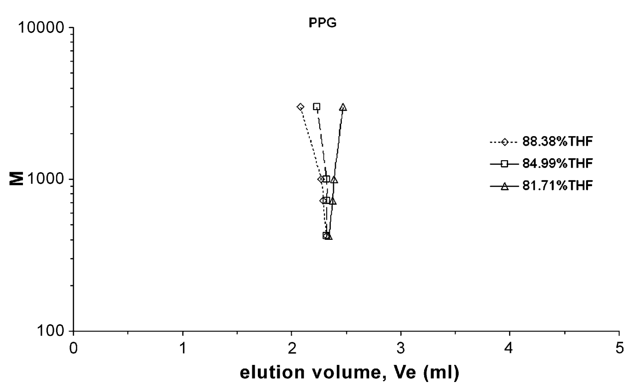

Fig. 4. Critical conditions for PPG on the Prodigy ODS3 column in THF–water.

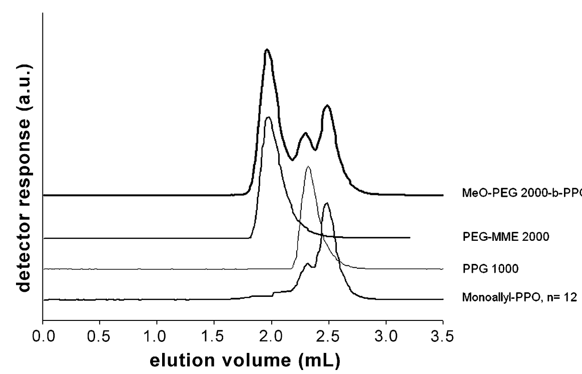

Fig. 5. LCCC of a diblock copolymer and the corresponding homopolymers on Prodigy ODS3 in 84.99 wt% THF (CAP for PPO); detection: ELSD.

the PPOs with allylic end group elute later than the PPGs, which coincide with the solvent peak.

In the paper mentioned above we have used an analytical column (Jordi Gel DVB 500 RP) packed with a stationary phase consisting of 100% poly(divinyl benzene). For this study, we wanted to perform the same separation also on a semipreparative scale. As a semipreparative column with the same chemical nature is not available, we have tried to find a CAP for PPO on an ODS column. As can be seen in Fig. 4, PPGs elute sufficiently close to critical conditions at a range between 81 and 85 wt% THF. In such a mobile phase, a separation of the block copolymer from PPGs and PPOs with allylic end group can be achieved (Fig. 5).

A similar separation is also possible on a semipreparative column: even with a rather large sample loop, the individual peaks are sufficiently resolved (Fig. 6). The fraction limits are indicated by dotted lines.

The individual fractions from LCCC were collected and the mobile phase was evaporated by high vacuum freeze drying below −50 °C with vacuum around 3 Pa. The dried fractions were subsequently analyzed by SEC, LAC and MALDI-TOF-MS.

4.1. Size exclusion chromatography in the second dimension

Fig. 7 shows an overlay of the chromatograms obtained from SEC for the raw sample and the individual fractions by using only density detection. Evidently, the copolymer (fraction 1) elutes earlier than the homopolymer fractions, as could be expected. The maximum of fractionated block copolymers is also somewhat earlier than the raw sample. Fraction 3 contains the lower molecular fraction, which should be the homopolymer, and fraction 2 seems to contain homo- and copolymer.

If this density detector is combined with some other concentration detector (like RI in this case), chemical composition across the MMD can be obtained.

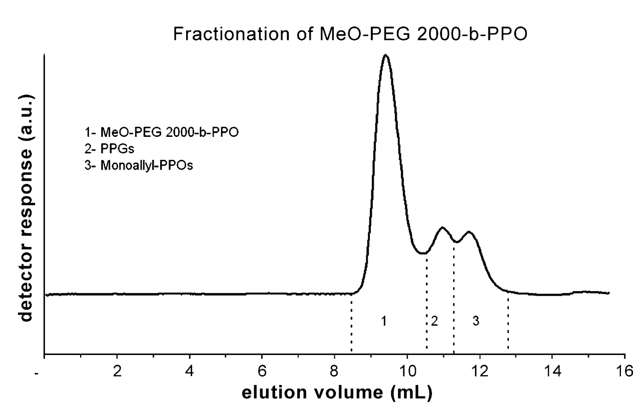

Fig. 6. LCCC of a diblock copolymer (MeO-PEG 2000-b-PPO) on Spherisorb ODS2 (semipreparative) in 81.14 wt% THF (CAP of PPO); detection: density.

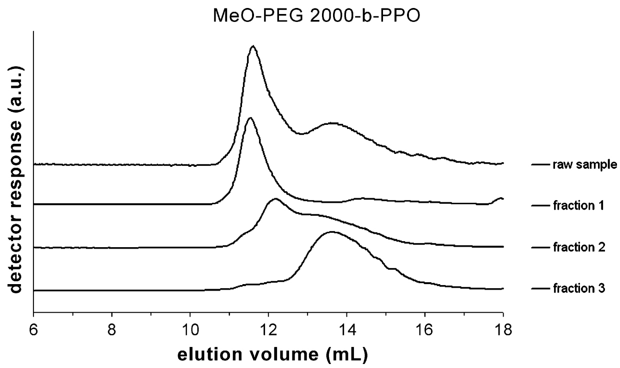

Fig. 7. Overlay of SEC chromatograms of MeO-PEG 2000-b-PPO and the fractions from LCCC (see Fig. 6); detection: density.

As can be seen in Fig. 8, the MMD of the copolymer (fraction 1) has a maximum at about 3000 g/mol, and the PO content of this fraction is somewhat less than 40%. The second maximum in Fig. 8 should be due to incomplete separation of the peaks in LCCC.

Figs. 9 and 10 show the results obtained for the other fractions: the MMD of fraction 2 has a maximum at about 2000. This fraction contains, however, close to 100% PO, which should correspond to the PPGs. Fraction 2 has a trimodal distribution: it obviously contains PPGs of different molar mass and also some block copolymer and maybe some PPO chains with allylic end group.

Fraction 3 (the PPOs with allylic end groups) has indeed a much lower molar mass (with a maximum of about 800). Based on retention time, this fraction could also be the initiator (PEG-MME). This can, however, be ruled out by dual detection, which shows, that fraction 3 consists also of 100% PO.

The small amount of block copolymer in fractions 2 and 3 may be due to an incomplete separation of the individual peaks in LCCC.

4.2. MALDI-TOF-MS in the second dimension

To confirm the chromatographic results, MALDI-TOF mass spectra were recorded. The raw sample gave a complex spectrum that is not shown here. Clearer spectra were obtained for the individual fractions 1–3 obtained from LCCC (see Fig. 6).

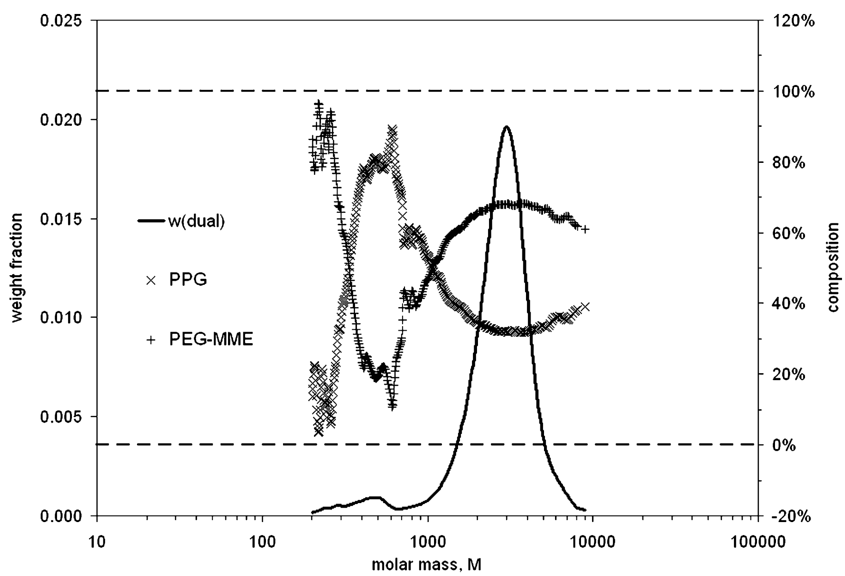

Fig. 8. MMD and chemical composition of fraction 1 from LCCC (see Fig. 6) containing the block copolymer, as obtained from SEC with dual detection.

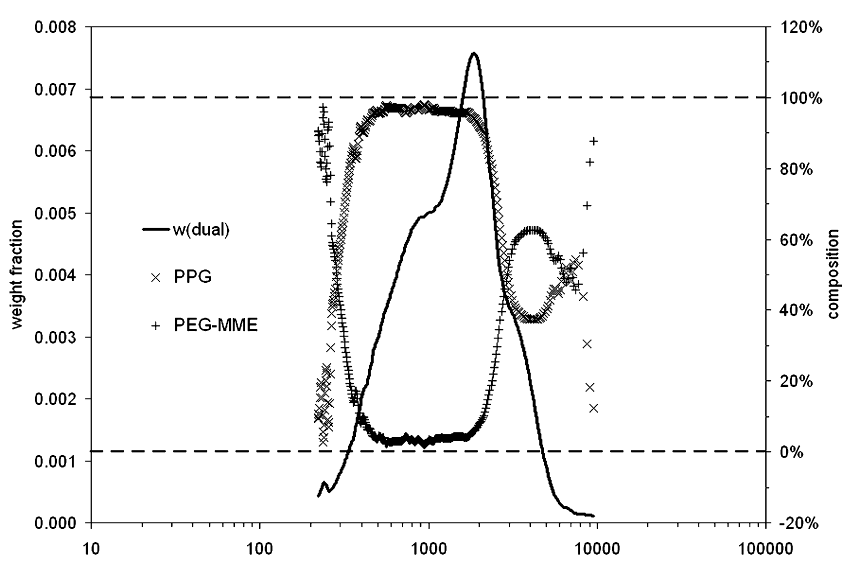

Fig. 9. MMD and chemical composition of fraction 2 from LCCC (see Fig. 6), presumably PPGs, as obtained from SEC with dual detection.

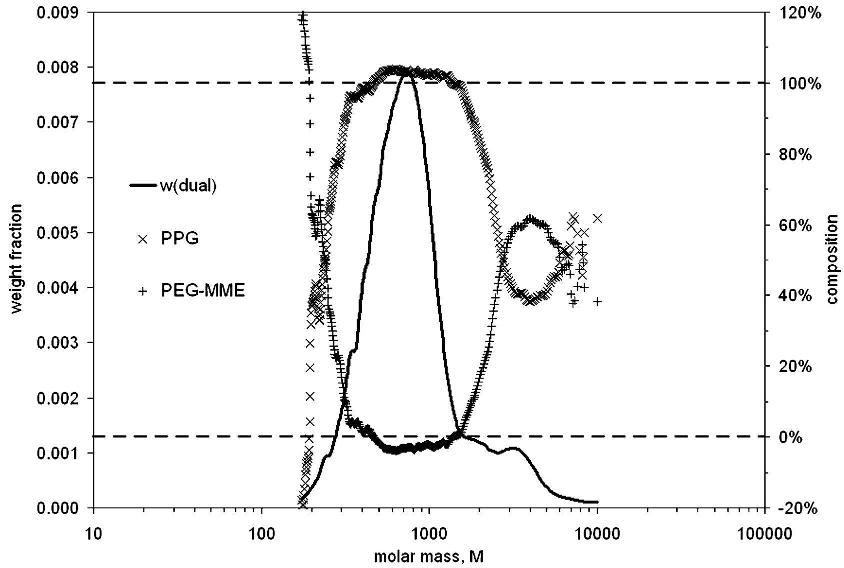

Fig. 10. MMD and chemical composition of fraction 3 from LCCC (see Fig. 6), presumably PPOs with allylic end group, as obtained from SEC with dual detection.

The MALDI spectrum of fraction 1 is reproduced in Fig. 11. The maximum of the EO–PO block copolymers was observed at about 2850 Da. Additionally, PPG was found with a maximum at about 1700–1750 Da. Keeping aspects like mass discrimination in MALDI-MS and polydispersity of the EO–PO block copolymers (in the EO as well as in the PO block) in mind, these results agree to those from chromatography.

Interesting results were found for fraction 2. As shown in Fig. 12, the fraction contained – as expected – PPGs with a maximum at approx. 1700 Da, as well as PPOs with allylic end group with a maximum around 750 Da. But it is important to note, that the MMD of the PPGs clearly expands into the low mass range 400–1000 Da (compare also inset in Fig. 12). For a typical MMD of the PPGs with the maximum at 1700 Da one would expect the low mass end of the MMD at about 1000 Da. Thus, the MALDI data indicate that ‘a second’ PPGs with a maximum of the MMD around 750 Da is also formed during the synthesis.

A possible explanation could be as follows: the higher molecular fraction of PPGs is formed with traces of water present in the initial polymerization mixture. As these chains grow on both ends for the entire reaction time, they will be quite large.

The lower molecular fraction obviously had less time to grow, which means, that it consists of chains started at high conversion. This could be due to the reaction shown in Fig. 2, which forms the anion of propylene glycol and thus starts a new chain.

After all, these MS results agree to those from chromatography, too.

The MALDI-TOF mass spectrum of fraction 3 is shown in Fig. 13. Fraction 3 (the PPOs with allylic end group) had indeed a lower molar mass with a maximum of about 700 Da and was pure. This is also reasonable: the PPO chains with allylic end groups formed by chain transfer reaction to monomer (Fig. 1) must be smaller, as this reaction occurs during the entire course of the polymerization.

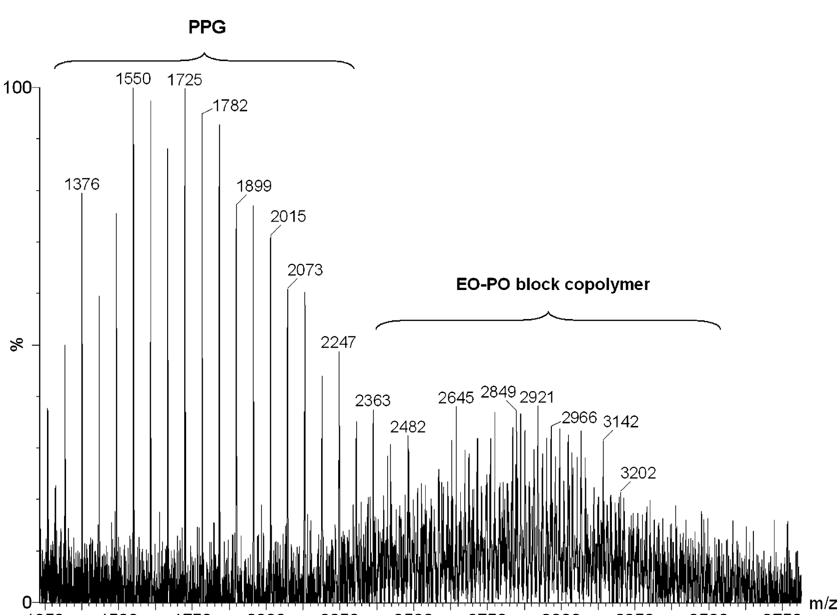

Fig. 11. MALDI-TOF mass spectrum of fraction 1 from LCCC of MeO-PEG 2000-b-PPO (see Fig. 6) showing ions $[MNa]^+$ of PPGs (e.g. $[HO(C_3H_6O)_{29}HNa]^+$ at $m/z = 1725$ Da) and EO–PO block copolymer (compare text).

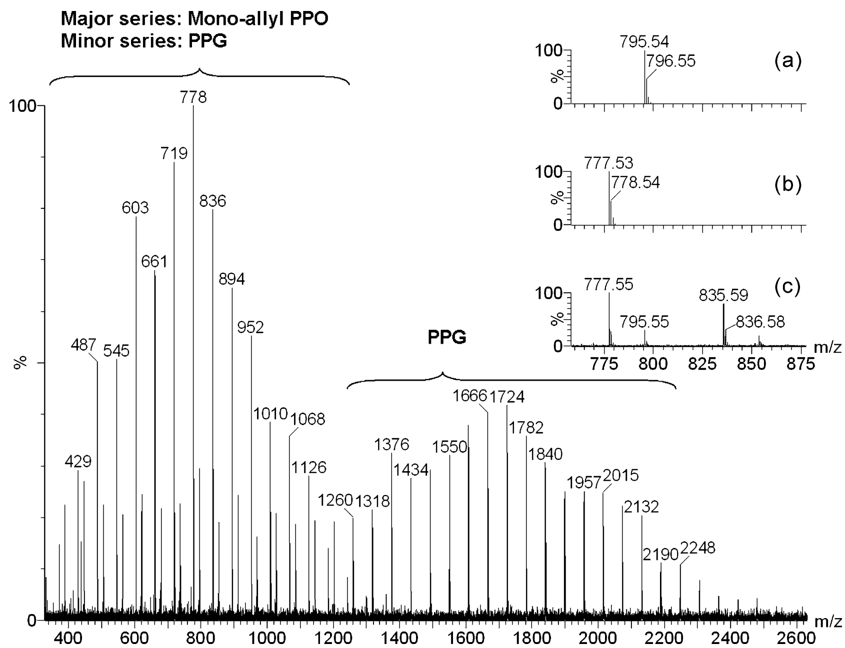

Fig. 12. MALDI-TOF-MS of fraction 2 from LCCC of MeO-PEG 2000-b-PPO (see Fig. 6). Inset: comparison of the theoretical isotope pattern of sodiated PPG ($n = 13$) (a), monoallyl-PPO ($n = 13$) (b) and the experimental result (c).

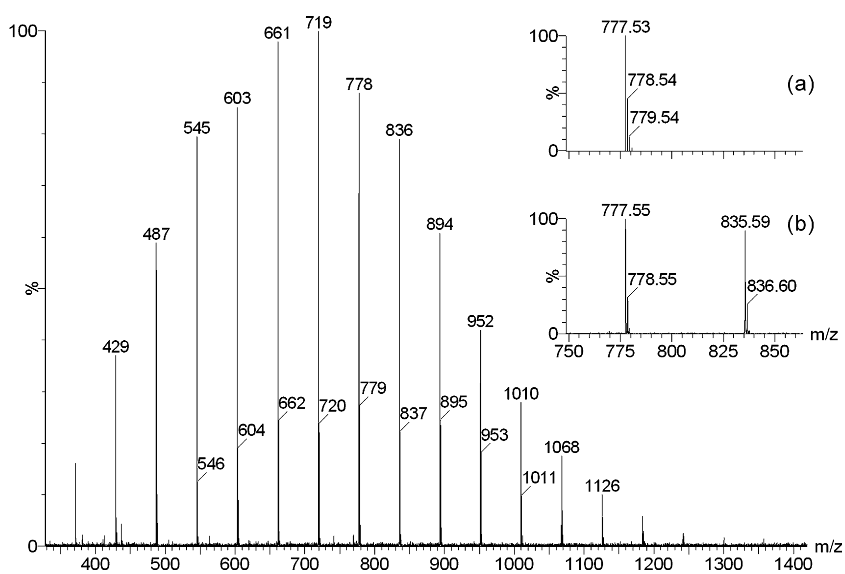

Fig. 13. MALDI-TOF-MS of fraction 3 from LCCC of MeO-PEG 2000-b-PPO (see Fig. 6). Inset: comparison of the theoretical isotope pattern of sodiated monoallyl-PPO ($n = 13$) (a) and the experimental result (b).

In priciple, PPO chains with two allylic end groups could also be formed, if a growing chain with an allylic end group (from chain transfer to the monomer, Fig. 1) would be terminated by chain transfer to the penultimate unit (Fig. 2). If one considers, that typically less than 10% of the growing chains are allyl PPO, and only a few of them will be terminated by chain transfer to the penultimate unit, one may expect less than 1% of diallyl terminated chains. Hence it is not surprising, that they could not be detected by any of the methods.

4.3. Liquid adsorption chromatography in the second dimension

As the separation power of SEC is limited, it appears attractive to use LAC in second dimension. This is rather easy in the case of homopolymers, as can be seen in Fig. 14, which shows a gradient separation of a mixture of three PPG samples with different molar

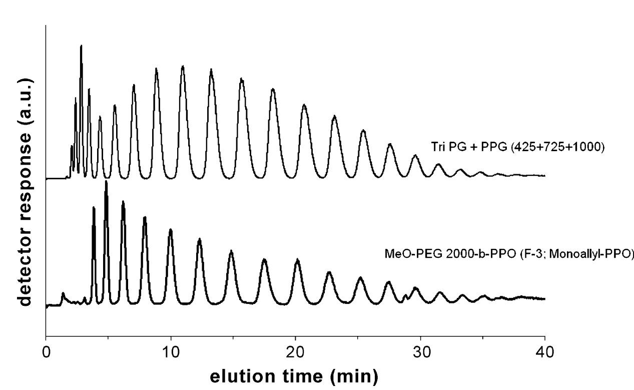

Fig. 14. Gradient LAC of a mixture of PPGs and fraction 3 from LCCC (see Fig. 6). Symmetry C18 (gradient: from 70 to 90 wt% methanol in 35 min); detection: ELSD.

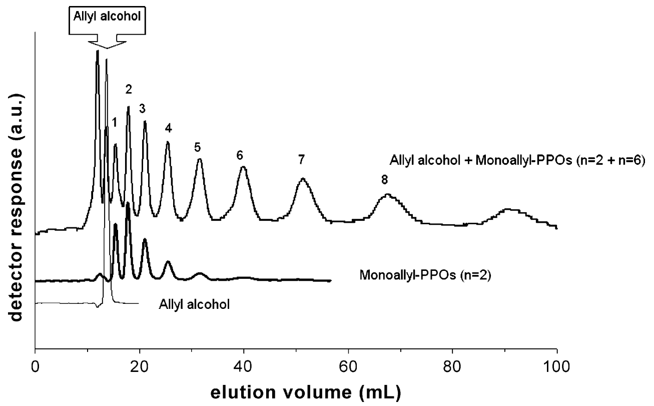

Fig. 15. LAC of allyl alcohol and monoallyl-PPOs with different average number of repeat units on the Spherisorb S5 ODS2 (semipreparative) column in methanol:water (70 wt%); detection: density.

mass (425, 725, and 1000), in which all oligomers are resolved to the baseline. A similar separation is also achieved with fraction 3 (from Fig. 6) from LCCC, which contains the PPOs with allylic end group. As the allyl end group is also adsorbed, the elution volumes of the individual oligomers should be higher than that of the glycols. Hence it was necessary to identify the individual peaks. This cannot be done by using allyl alcohol as internal standard, as the alcohol is not detected by the ELSD.

We have thus fractionated some samples of PPOs with allylic end group, which had been prepared by propoxylation of allyl alcohol [7].

On the sempreparative Spherisorb ODS2 column, a reasonable separation of the lower oligomers can be achieved, as can be seen in Fig. 15.

The fractions from this separation were used as internal standards in the gradient separation. Fig. 16 shows two chromatograms of fraction 3 from LCCC, one of which is spiked with the oligomer with 6 PO units (from Fig. 15).

The next question concerned the separation of the block copolymer fraction. In principle, a LAC separation according to the number of PO units would require critical conditions for the EO block. There is a CAP for EO at about 80 wt% methanol on ODS columns, which is, however, rather a plateau [41]. On typical C18 columns, the elution volumes of lower PEGs in mobile phases containing 70–95 wt% methanol are almost independent on molar mass [41]. Hence we tried to run a gradient separation in a range close to the CAP for EO: as can be seen in Fig. 17, the chromatogram of the copolymer fraction (F1) also shows sharp peaks, which are resolved to the baseline. A comparison with the chromatograms of the raw sample and the fraction containing the PPOs with allylic end group shows, that the PO block in the block copolymer is indeed longer than the allylic PPO chain.

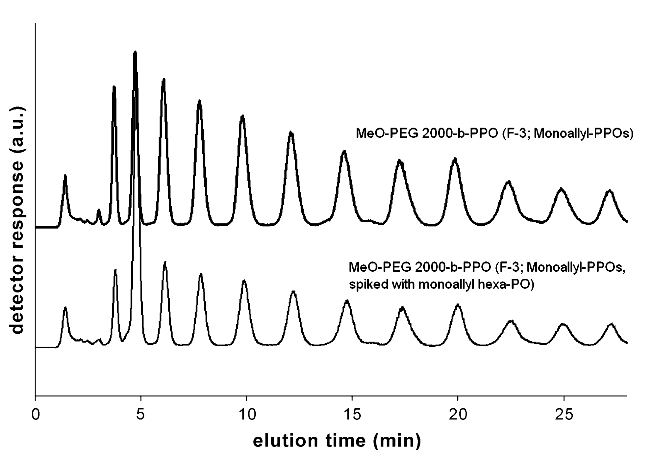

Fig. 16. LAC of MeO-PEG 2000-b-PPO (fraction 3, see Fig. 6) spiked with monoallyl hexa(propylene oxide) (from the preparative separation shown in Fig. 15); conditions as in Fig. 14.

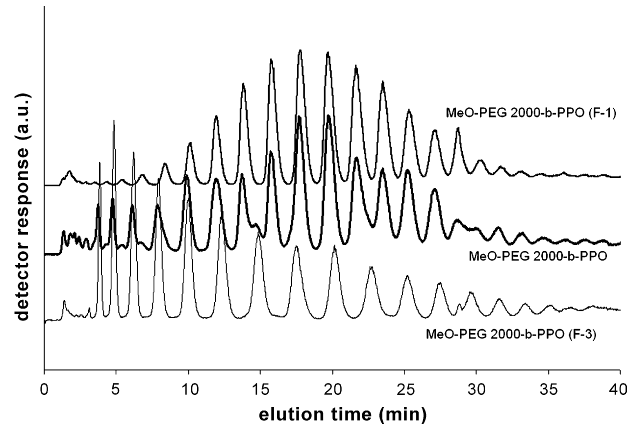

Fig. 17. LAC of MeO-PEG 2000-b-PPO (raw sample), and the fractions 1 and 3 from LCCC (see Fig. 6). Conditions as in Fig. 14.

5. Conclusion

A complete characterization of EO–PO block copolymers can be achieved using combinations of different chromatographic techniques and MALDI-TOF-MS. Each of these methods, when applied alone, cannot provide all of the required information. As a first and rapid method, SEC with dual detection shows the overall MMD and the chemical composition along the MMD. LCCC reveals the functionality and allows to separate the block copolymer from the homopolymers. MALDI-TOF-MS is difficult to interprete, when applied to the raw samples, but yields excellent results, when applied to fractions from LCCC. These fractions can also be analyzed by LAC, which allows a full separation of oligomers. The results obtained with the different techniques are in excellent agreement. This could be very useful for optimizing synthesis parameters and physical properties of these important materials.

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